Chapter 18, Problem 9PP

Problem

(a) Calculate the total energy absorbed by a 1-**Ω** resistor with *i*(*t*) = 10*e*−2|*t*| A in the time domain.

(b) Repeat (a) in the frequency domain.

Step-by-step solution

Step 1 of 8

(a)

Consider the following function.

$$i(t) = 10e^{-2|t|} \text{ A}$$

Find the value of the total energy absorbed by the $1-\Omega$ resistor.

Write the expression for the total energy absorbed by the $1-\Omega$ resistor in time domain.

$$\begin{aligned} W_{1-\Omega} &= \int_{-\infty}^{\infty} i^2(t)\,dt \\ &= \int_{-\infty}^{\infty} \left[10e^{-2|t|}\right]^2 dt \\ &= \int_{-\infty}^{\infty} \left[100e^{-4|t|}\right] dt \\ &= \int_{-\infty}^{0} \left[100e^{-4|t|}\right] dt + \int_{0}^{\infty} \left[100e^{-4|t|}\right] dt \end{aligned}$$

Step 2 of 8

Simplify $W_{1-\Omega}$ further.

$$\begin{aligned} W_{1-\Omega} &= 100\int_{-\infty}^{0} \left[e^{-4(-t)}\right] dt + 100\int_{0}^{\infty} \left[e^{-4t}\right] dt \\ &= 100\left[\frac{e^{4t}}{4}\right]_{-\infty}^{0} + 100\left[\frac{e^{-4t}}{-4}\right]_{0}^{\infty} \\ &= 100\left[\frac{e^{4(0)} - e^{4(-\infty)}}{4}\right] + 100\left[\frac{e^{-4(\infty)} - e^{-4(0)}}{-4}\right] \\ &= 100\left[\frac{1-0}{4}\right] + 100\left[\frac{0-1}{-4}\right] \end{aligned}$$

Step 3 of 8

Simplify $W_{1-\Omega}$ further.

$$\begin{aligned} W_{1-\Omega} &= 100\left[\frac{1}{4}\right] + 100\left[\frac{1}{4}\right] \\ &= \frac{200}{4} \\ &= 50 \text{ J} \end{aligned}$$

Therefore, the required total energy absorbed by the $1-\Omega$ resistor is $\boxed{50 \text{ J}}$.

Step 4 of 8

(b)

Find the Fourier transform of the expression $i(t)$.

Apply Fourier transform to the expression $i(t)$.

$$\mathcal{F}[i(t)] = \mathcal{F}\left[10e^{-2|t|}\right]$$

$$\begin{aligned} I(\omega) &= 10\mathcal{F}\left[e^{-2|t|}\right] \\ &= 10\left[\frac{2(2)}{2^2+\omega^2}\right] \quad \left[\text{Since, } \mathcal{F}\left[e^{-a|t|}\right] = 10\left[\frac{2(a)}{a^2+\omega^2}\right]\right] \\ &= \frac{40}{(4+\omega^2)} \end{aligned}$$

Therefore, the Fourier transform of the expression $i(t)$ is $\frac{40}{(4+\omega^2)}$.

Step 5 of 8

Find the value of the total energy absorbed by the $1-\Omega$ resistor in Frequency domain.

Write the expression for the total energy absorbed by the $1-\Omega$ resistor in Frequency domain.

$$\begin{aligned} W_{1-\Omega} &= \frac{1}{2\pi}\int_{-\infty}^{\infty} |I(\omega)|^2 \, d\omega \\ &= \frac{1}{2\pi}\int_{-\infty}^{\infty} \left[I(\omega) I^*(\omega)\right] d\omega \\ &= \frac{1}{2\pi}\int_{-\infty}^{\infty} \left[\frac{40}{(4+\omega^2)} \frac{40}{(4+\omega^2)}\right] d\omega \\ &= \frac{1}{2\pi}\int_{-\infty}^{\infty} \left[\frac{1,600}{(4+\omega^2)^2}\right] d\omega \end{aligned}$$

Step 6 of 8

Simplify $W_{1-\Omega}$ further.

$$W_{1-\Omega} = \frac{800}{\pi}\int_{-\infty}^{\infty} \left[\frac{1}{(4+\omega^2)^2}\right] d\omega$$

Substitute $\tan\theta$ for ω in this expression.

$$\begin{aligned} W_{1-\Omega} &= \frac{800}{\pi}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left[\frac{1}{(4+4\tan^2\theta)^2}\right] (2\sec^2\theta d\theta) \\ &= \frac{1,600}{\pi}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left[\frac{1}{16(1+\tan^2\theta)^2}\right] (\sec^2\theta d\theta) \\ &= \frac{100}{\pi}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left[\frac{1}{(\sec^2\theta)^2}\right] (\sec^2\theta d\theta) \end{aligned}$$

Step 7 of 8

Simplify $W_{1-\Omega}$ further.

$$W_{1-\Omega} = \frac{100}{\pi}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left[\frac{1}{\left(\sec^2\theta\right)}\right](d\theta)$$
$$= \frac{100}{\pi}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left[\cos^2\theta\right](d\theta)$$
$$= \frac{50}{\pi}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left[1+\cos 2\theta\right]d\theta$$
$$= \frac{50}{\pi}\left\{\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}[1]d\theta + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}[\cos 2\theta]d\theta\right\}$$

Step 8 of 8

Simplify $W_{1-\Omega}$ further.

$$W_{1-\Omega} = \frac{50}{\pi}\left\{\left[\frac{\pi}{2}-\left(-\frac{\pi}{2}\right)\right]+\left[\frac{\sin 2\left(\frac{\pi}{2}\right)-\sin 2\left(\frac{-\pi}{2}\right)}{2}\right]\right\}$$
$$= \frac{50}{\pi}\left\{\pi+\left[\frac{\sin\pi+\sin\pi}{2}\right]\right\}$$
$$= \frac{50}{\pi}\left\{\pi+\left[\frac{0+0}{2}\right]\right\}$$
$$= \frac{50}{\pi}\{\pi\}$$
$$W_{1-\Omega} = 50\text{ J}$$

Therefore, the value of the total energy absorbed by the $1-\Omega$ resistor in Frequency domain is $\boxed{50\text{ J}}$.